

NAME = PRATHAM 🏠

USN = 2VX24UE039

EXPT = 09

TITLE = Building logic into Improve Programming Skills

(i) Write a Python program to check whether a number is positive, negative or zero

```
In [16]: 1 num = float(input("Enter a number: "))
2
3 if num > 0:
4     print("The number is positive.")
5 elif num < 0:
6     print("the number is negative.")
7 else:
8     print("The number is zero.")
```

Enter a number: 7
The number is positive.

(ii) Write a Python program to find a quadrant in which the number x and y lie

```
In [17]: 1 x = float(input("Enter x-coordinate:"))
2 y = float(input("Enter y-coordinate:"))
3
4 if x>0 and y>0:
5     print("The point lies in First Quadrant")
6 elif x<0 and y>0:
7     print("The point lies in Second Quadrant")
8 elif x<0 and y<0:
9     print("The point lies in Third Quadrant")
10 elif x>0 and y<0:
11     print("The point lies in Fourth Quadrant")
12 elif x== 0 and y== 0:
13     print("The point lies at the Origin")
14 elif x== 0:
15     print("The point lies on Y-axis")
16 else:
17     print("The point lies on X-axis")
```

Enter x-coordinate:4
Enter y-coordinate:7
The point lies in First Quadrant

(iii) Write a Python program to check whether a number is even or odd

```
In [18]: 1 num = int(input("Enter a number: "))
2
3 # Check even or odd
4 if num % 2 == 0:
5     print("The number is Even")
6 else:
7     print("The number is Odd")
```

Enter a number: 4
The number is Even

(iv) Write a Python program to print the following patterns:

(a)

```
* * *
* * *
* * *
```

In [19]:

```
1
2 for i in range(1,5):
3     for j in range(1,4):
4         print(" *", end=" ")
5     print()
```

```
* * *
* * *
* * *
* * *
```

(b)

```
1 1 1
2 2 2
3 3 3
```

In [9]:

```
1 rows = 3
2 cols = 3
3
4 for i in range(1, rows + 1):
5     for j in range(cols):
6         print(i, end=" ")
7     print()
```

```
1 1 1
2 2 2
3 3 3
```

(c)

```
5 6 7
5 6 7
5 6 7
5 6 7
```

In [7]:

```
1 rows = 4
2 start = 5
3 cols = 3
4
5 for i in range(rows):
6     print(" ", end=" ") # for leading space (indent)
7     for j in range(start, start + cols):
8         print(j, end=" ")
9     print()
```

```
5 6 7
5 6 7
5 6 7
5 6 7
```

(d)

```
NAME
NAME NAME
NAME NAME NAME
NAME NAME NAME NAME
```

```
In [10]: 1 rows = 4
2
3 for i in range(1, rows + 1):
4     for j in range(i):
5         print("NAME", end=" ")
6     print()
```

```
NAME
NAME NAME
NAME NAME NAME
NAME NAME NAME NAME
```

(V) Write the Python program to print the following sequence of numbers using loop:

```
7
14
21
28
35
42
49
```

```
In [15]: 1
2 for i in range(1,8):
3     print(i*7)
4
```

```
7
14
21
28
35
42
49
```

```
In [ ]: 1
```